Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science &Engineering

Course Code: UCS510, UCS319 Course Name: Computer Architecture and Organization	EST: Summer Sem. 2025
19/07/2025	9:00 AM - 12:00 PM
Time: 3 Hours, M. Marks: 40	Faculty: Dr. Yadwinder Singh

Note: Attempt any five questions in a proper sequence. Assume missing data, if any, suitably.

Q.1. **(7+1)**

A. Consider a 16-bit processor in which the following appears in main memory, starting at location 200:

200	Load to AC	Mode
201	500	
202	Next Instruction	

The first part of the first word indicates that this instruction loads a value into an accumulator. The Mode field specifies an addressing mode and, if appropriate, indicates a source register; assume that when used, the source register is R1, which has a value of 400. There is also a base register that contains the value 100. The value of 500 in location 201 may be part of the address calculation. Assume that location 399 contains the value 999, location 400 contains the value 1000, and so on. Determine the effective address and the operand to be loaded for the following address modes:

a) Direct b) Immediate c) Indirect	d) PC Relative e) Register f) Register Indirect	g) Autoincrement using R1

B. A computer has 32-bit instructions and 12-bit addresses. If there are 250 two-address instructions, how many one-address instructions can be formulated?

Q.2 **(2+4+2)**

A. Represent the following decimal numbers in both binary sign/magnitude and twos complement using 16 bits: a) +512 b) -29

B. Assume numbers are represented in 8-bit twos complement representation. Show the calculation of the following decimal numbers (using specified representation), give decimal representation of the final calculated result.
a) 6 +13 b) -6 + 13 c) 6 – 13 d) -6 -13

C. Differentiate between Interrupted I/O and Programmed I/O. (Give 4 differences only)

Q.3	A. What are different pipeline hazards? Discuss briefly with example. B. Explain the detailed working of DMA Controller with a neat diagram.	**(4+4)**
Q.4	A. Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate: - 1. Pipeline cycle time 2. Non-pipeline execution time 3. Speed up ratio 4. Pipeline time for 1000 tasks 5. Sequential time for 1000 tasks 6. Throughput B. A cache is having 60% hit ratio for read operation. Cache access time is 30 ns and main memory access time is 100 ns, 50% operations are read operation. What will be the average access time for read operation?	**(6+2)**
Q.5	A. A digital computer has a memory of 64K x 16 and a cache memory of 1K words. The cache uses direct mapping with a block size of four words. a) How many bits are there in the tag, index, block and word fields of the address format? b) How many bits are there in each word of cache and what is the size of cache? c) How many blocks can the cache accommodate? B. a) How many 128 X 8 RAM chips are needed to provide the memory capacity of 2048 bytes? b) How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips? c) How many lines must be decoded for chip select? Specify the size of the decoders.	**(5+3)**
Q.6	Write a program to evaluate the arithmetic statement: $$X = \frac{A - B + C * (D * E - F)}{G + H * K}$$ a) Using a general register computer with three address instructions. b) Using a general register computer with two address instructions. c) Using an accumulator type computer with one address instructions. d) Using a stack organized computer with zero-address instructions.	**(8)**